A. Greatest Convex

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, math, number theory | https://codeforces.com/problemset/problem/1768/A

You are given an integer k. Find the largest integer x, where $1 \le x$ lt k, such that $x! + (x - 1)!^{\text{dagger}}$ is a multiple of ^{ddagger} k, or determine that no such x exists.

^{dagger} $y!$ denotes the factorial of y, which is defined recursively as $y! = y \cdot (y\text{-}1)!$ for $y \ge 1$ with the base case of $0! = 1$. For example, $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 0! = 120$.

^{ddagger} If a and b are integers, then a is a multiple of b if there exists an integer c such that $a = b \cdot c$. For example, 10 is a multiple of 5 but 9 is not a multiple of 6.

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of test cases follows.

The only line of each test case contains a single integer k ($2 \le k \le 10^9$).

Output

For each test case output a single integer — the largest possible integer x that satisfies the conditions above.

If no such x exists, output -1.

Example

Input
4 3 6 8 10

Output
2 5 7 9

Note

In the first test case, $2! + 1! = 2 + 1 = 3$, which is a multiple of 3.

In the third test case, $7! + 6! = 5040 + 720 = 5760$, which is a multiple of 8.